

Validation of the Fast Calorimeter Simulation in LHCb Fixed-Target Collisions

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June 12, 2026

- 1 Experimental context: LHCb, SMOG2, ECAL
- 2 What is **fast simulation**? Why?
- 3 Simulation samples and validation strategy
- 4 Baseline results in pH_2
- 5 Extension to other fixed-target systems
- 6 Conclusions and next steps

Main question

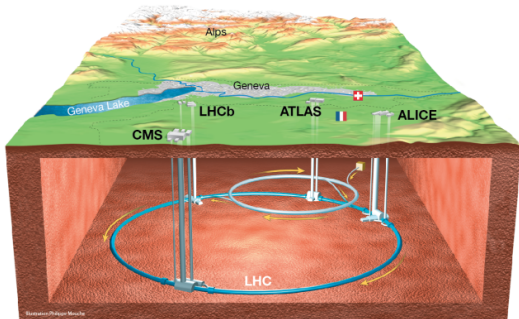
Can the fast calorimeter simulation reproduce the detailed simulation for photon-related observables in fixed-target χ_c samples?

How to answer?

- Compare shapes, not only single numbers.
- Look for variable-, energy-, and region-dependent effects.

Experimental context: LHCb at the LHC

The LHC is CERN's high-energy proton–proton collider, designed to produce rare particles and precision measurements that test the Standard Model and search for new physics.



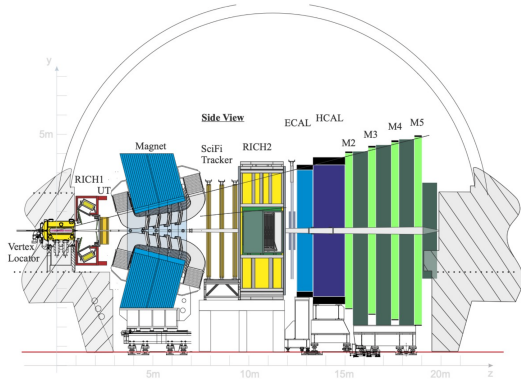
LHC ring and the main experiments.

- LHCb is a **single-arm forward spectrometer**.
- Optimized for heavy-flavour physics.
- Forward coverage: approximately $2 < \eta < 5$.
- Run 3 upgrade: full software trigger and higher data rate.

Why this matters

More data $\Rightarrow$ more simulated samples needed $\Rightarrow$ simulation speed becomes critical.

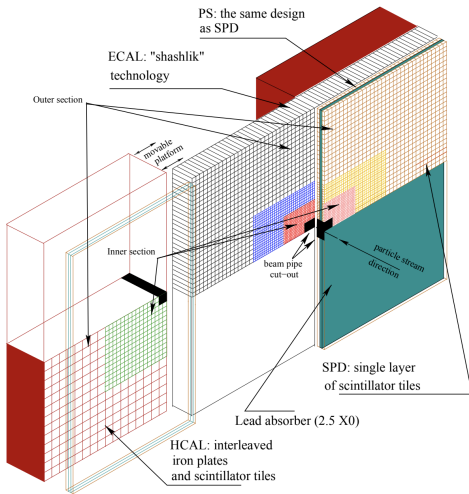
The LHCb detector and photon reconstruction



Layout of the upgraded LHCb detector.

- VELO + tracking stations (UT, SciFi) reconstruct the trajectory of charged particles.
- RICH + ECAL/HCAL detectors provide particle identification.
- ECAL is central for reconstructing photons and electrons by measuring their electromagnetic showers.
- Muon chambers reconstruct $J/\psi \rightarrow \mu^+ \mu^-$.

Calorimeter system: where this validation acts



LHCb calorimeter system.

ECAL regions

- Inner, middle, and outer regions.
- Cell size increases away from the beam pipe.
- Different occupancies and resolutions can affect photon response.

Validation target

Does fast simulation reproduce the ECAL response for photons in SMOG2 samples?

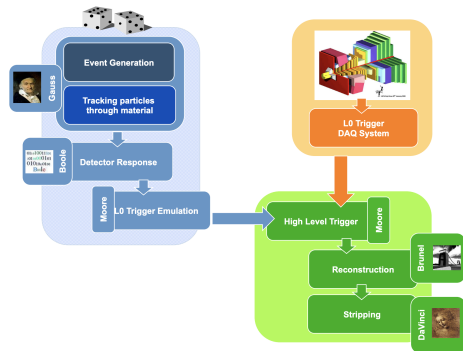
Why fast simulation?

Problem

- Physics analyses need large Monte Carlo samples.
- Full detector simulation is accurate but expensive.
- The calorimeter is one of the most CPU-heavy regions ($> 50\%$).

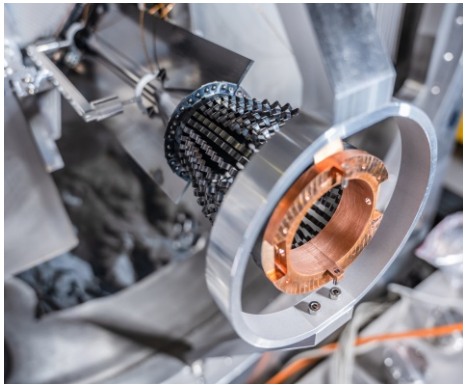
This project

- Validate the **fast calorimeter simulation** in SMOG2 fixed-target conditions.
- Focus on photons from $\chi_c \rightarrow (J/\psi \rightarrow \mu^+ \mu^-) \gamma$.



Monte Carlo and data-processing chain.

SMOG2: fixed-target collisions at LHCb



SMOG2 gas-storage cell installed upstream of the VELO.

Principle

- Gas is injected into a storage cell.
- The LHC proton beam collides with gas nuclei.
- LHCb records these as fixed-target collisions.

For this project

- SMOG2 supports several gases: H_2 , D_2 , He, Ne, Ar, Xe.
- Beam–gas and beam–beam data can be taken simultaneously.
- We test whether fast ECAL simulation still works in this geometry.

Signal channel

$$\chi_c \rightarrow J/\psi \gamma, \quad J/\psi \rightarrow \mu^+ \mu^-.$$

- J/ψ is reconstructed by two muons.
- Photon reconstruction reflects the χ_c mass resolution.
- Therefore the ECAL response is directly relevant.

Main variables

- Photon momentum: p, p_x, p_y, p_z, p_T
 - Angles: $\phi, \theta, \eta, y, y^*$
 - Photon energy
 - ECAL region
- } for the cuts
- Reconstructed χ_c mass
 - Photon identification variables:
 `photon_IS_NOT_H`

- 2024 fixed-target samples at $\sqrt{s_{NN}} = 70.9$ GeV.
- Collision systems:

pH_2 ppH_2 pAr $ppAr$ $PbAr$

- Fast simulation: about 10^6 events per system.
- Detailed simulation: about 10^7 events per system.

Detailed vs fast

- ① Same generated signal decay.
- ② Different calorimeter response model.
- ③ Compare reconstructed distributions after the same selections.

Truth matching: selecting the intended decay chain

- Truth matching helps to identify/select the true photon from χ_c decay
- Use TRUEID to match reconstructed particles to generated particles.
- Use MC_MOTHER_ID to check the decay chain.
- Default comparisons use truth matching unless stated otherwise.

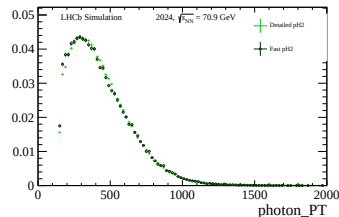
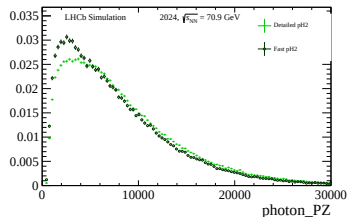
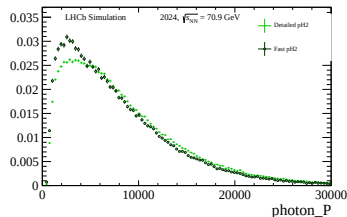
System	Total	Truth matched	Ratio
pH_2	19.45M	1.17M	6.02%
ppH_2	75.33M	1.04M	1.38%
pAr	39.62M	1.36M	3.44%
$ppAr$	51.38M	0.61M	1.18%
PbAr	16.33M	0.08M	0.46%

Number of candidates before and after the truth matching in detailed simulations.

Interpretation

Truth-matched samples isolate correctly reconstructed signal candidates, but also reduce statistics strongly.

Baseline system: p_{H_2} photon momentum



Photon momentum, longitudinal momentum, and transverse momentum after truth matching.

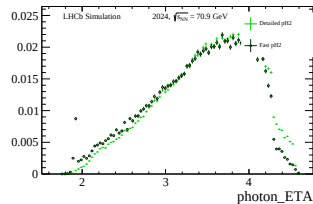
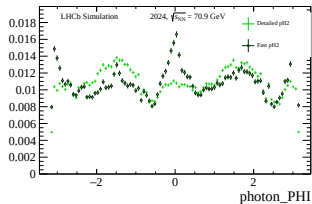
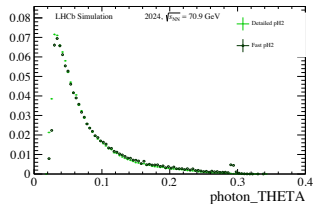
p and p_z

- Similar global shape.
- Main discrepancy at low momentum.
- Fast simulation overestimates in the low- p region.

p_T

- Overall good fit.

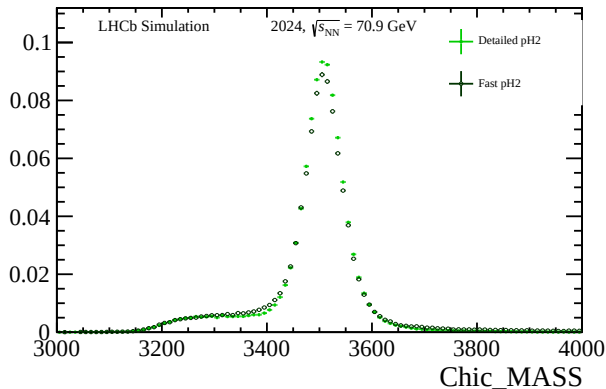
pH_2 : angular variables



polar angle θ , azimuthal angle ϕ , and pseudorapidity η .

- θ is mostly well, except for ~ 0.3 .
- Visible discrepancies in ϕ .
- η shows region-dependent differences.

Main physics observable: reconstructed χ_c mass



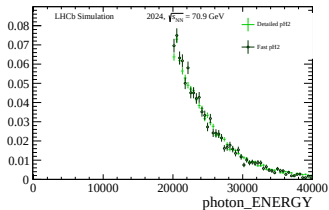
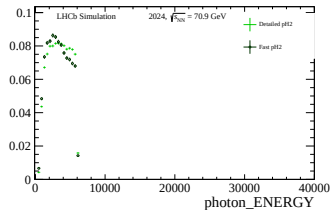
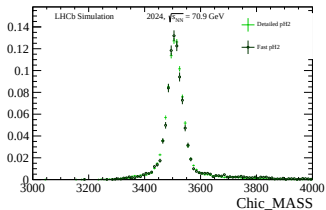
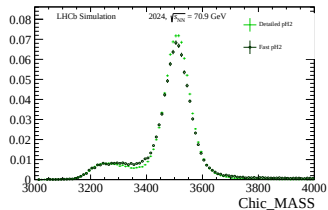
χ_c mass distribution in the baseline pH_2 sample.

Result

The overall mass shape is reproduced reasonably well.

- This is encouraging for χ_c analyses.
- However, there are still small discrepancies on reproducing the mass resolution.
- Investigation of this discrepancy is the natural next step.

Understanding the discrepancy: reconstructed χ_c mass and γ energy

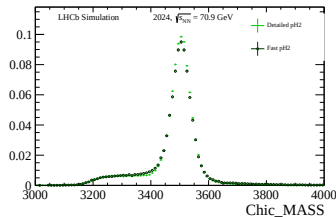


Key observation

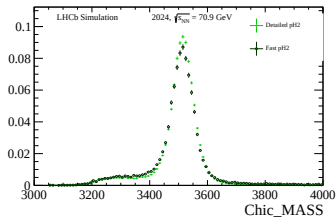
Low-energy photons show the largest visible discrepancies.

- Important because photon energy is highly related to χ_c mass.
- High-energy subset is generally better modeled.

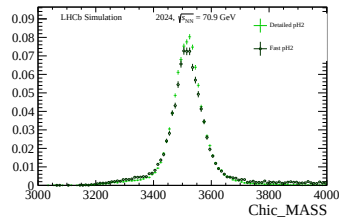
Understanding the discrepancy: reconstructed χ_c mass and ECAL regions



(a) inner



(b) middle

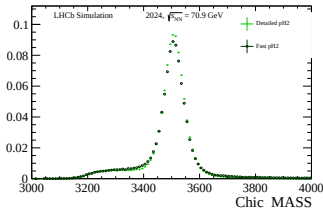
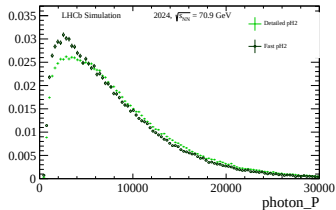
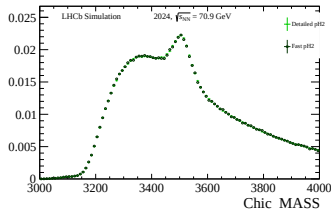
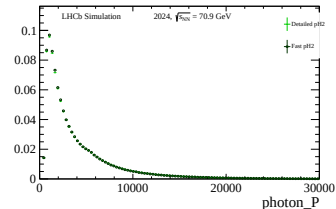


(c) outer

χ_c mass in inner, middle, and outer ECAL regions.

- The comparison depends on calorimeter region.
- Region effects are expected from different cell sizes and occupancies.
- Useful diagnostic for improving the fast model.

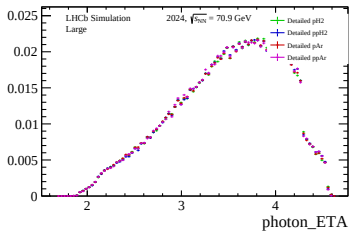
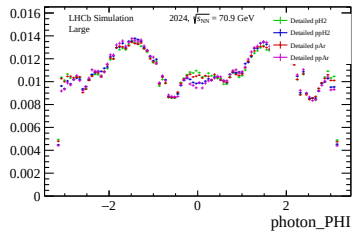
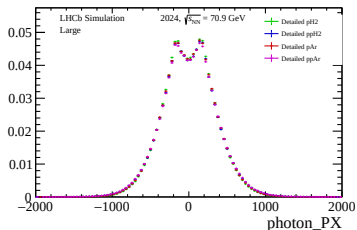
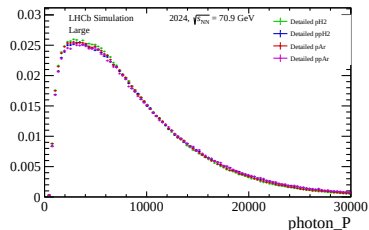
Effect of truth matching



Top: no truth matching. Bottom: truth matched.

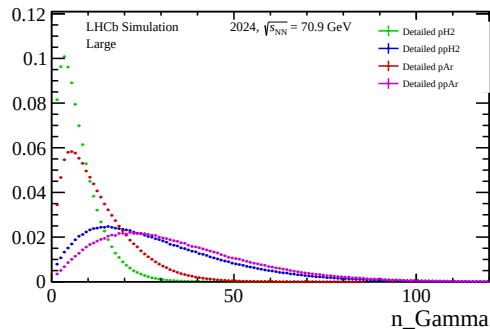
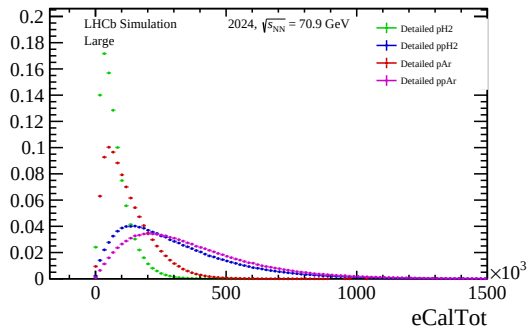
- Without truth matching, agreement is visibly better.
- With truth matching, discrepancies become more pronounced.
- The selection probes reconstruction-chain effects, not only raw calorimeter response.

Investigation of other SMOG2 systems (similarities)



- No significant differences in the photon kinematics.
- Slight noise in photon_PHI.

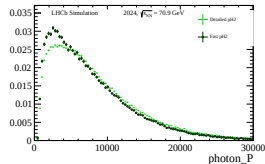
Investigation of other SMOG2 systems (differences)



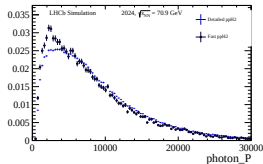
Total energy deposited in the ECAL, and the number of reconstructed photons.

Significant differences across systems.

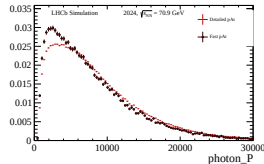
Extension of the Validation to other SMOG2 systems



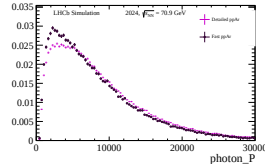
pH_2



ppH_2



pAr

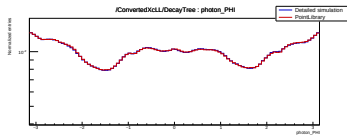
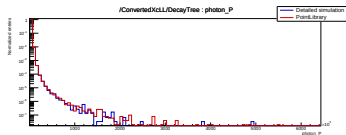
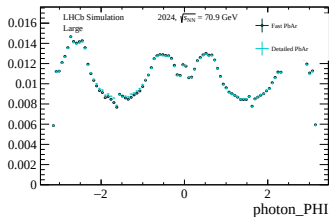
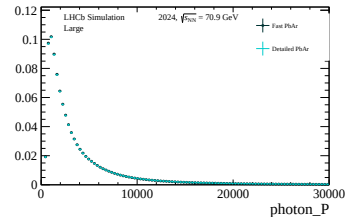


$ppAr$

Fast vs detailed comparison of photon_P across systems.

- Similar discrepancy patterns appear in all systems.
- Not specific to only pH_2 .
- Indicates a common photon/ECAL modeling effect.

Cross-check in PbAr collisions



- Comparison performed without truth matching.
- Good agreement is observed.
- Cross-check confirms consistency of the validation procedure.

Top: this validation. Bottom: cross-check plots from the development team.

Summary of observations

What works well

- Overall distribution shapes are often reproduced.
- Reconstructed χ_c mass is reasonably described.
- Similar behavior across several SMOG2 collision systems.

Main discrepancies

- Low-energy and low-momentum photons.
- Angular variables, especially ϕ and high/low η regions.
- Stronger differences after truth matching.

Fast simulation is promising, but not yet a black-box replacement for detailed simulation.

Conclusion

- The fast calorimeter simulation gives a reasonable first description of photon-related observables in SMOG2 χ_c samples.
- Non-negligible discrepancies remain and should be understood before precision use.
- All observations are reported to the fast simulation development team, and contributed towards the final result.

Next steps

- ① Add ratio plots and quantitative metrics.
- ② Study ECAL-region dependence in more detail.
- ③ Investigate the origin of truth-matching differences.
- ④ Validate for electrons from converted photons

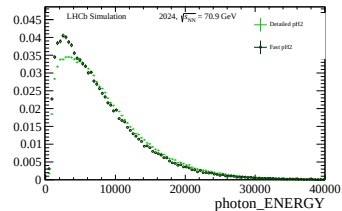
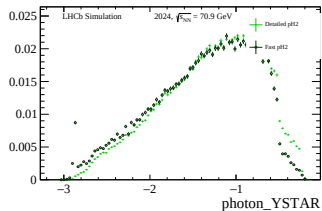
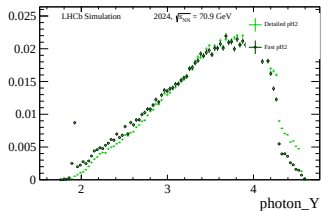
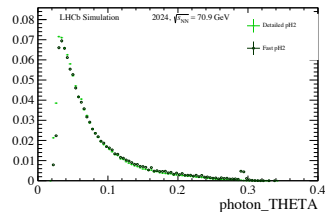
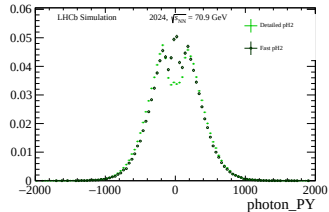
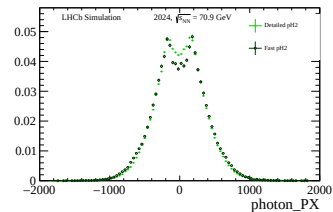
Thank you!

Questions?

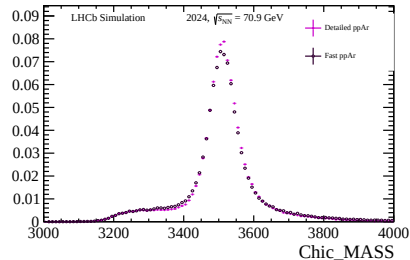
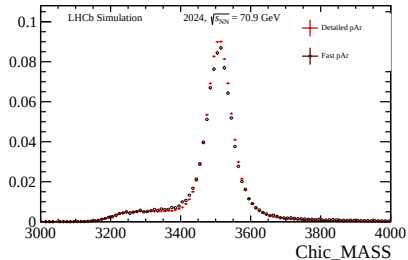
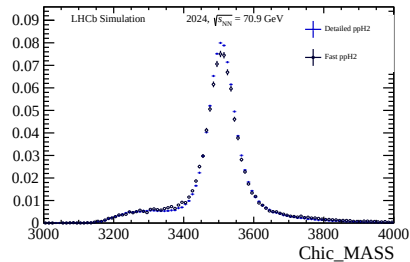
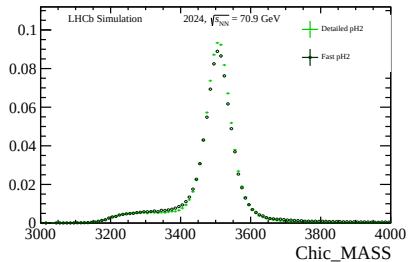
Take-home message

First experience on validating the fast calorimeter simulation on LHCb
Fixed Target collision: fast simulation is working quite well, but remaining discrepancies need further tunings.

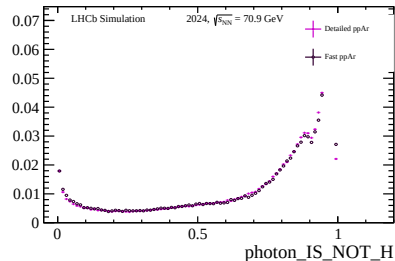
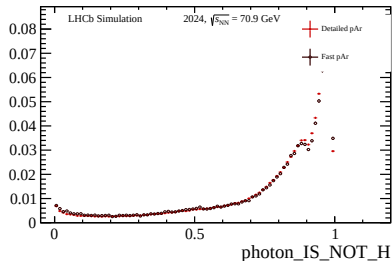
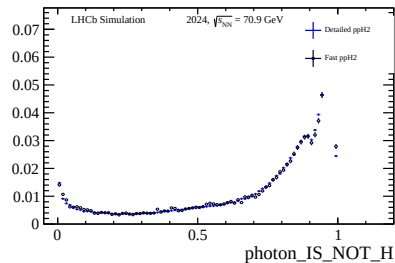
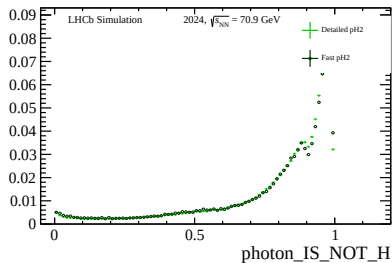
Backup: additional p_{H_2} variables



Backup: χ_c mass across collision systems

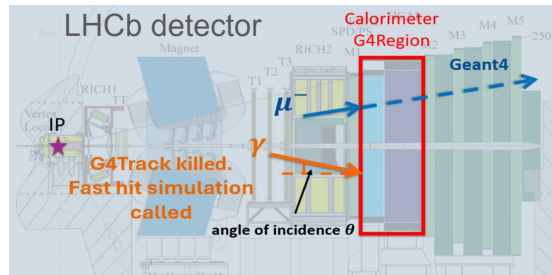


Backup: photon identification variables



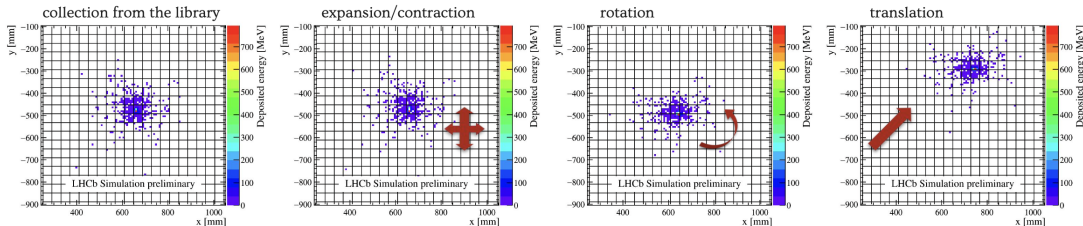
What is replaced? Why?

- At the calorimeter G4Region, a particle can be selected for fast simulation (e.g. a photon).
- The calorimeter response is then generated directly.
- Calorimeters dominate the detector-simulation CPU time.
- Avoiding full shower propagation gives a large speed gain.



Photon track is stopped at the calorimeter region and replaced by fast-generated hits.

Backup: PointLibrary method



Library construction

- 1 Full-simulate single particles at CALO entrance.
- 2 Use discrete nodes in momentum and angle: (p, θ) .
- 3 Store deposited-energy patterns as a library of “points”.

Event simulation

- Pick a stored point collection.
- Transform it according to the real particle:

$$p, \theta, \phi, x, y$$

- Apply scaling, rotation, and translation.